

Course outline (Syllabus of the course)

Course Code: CS1003 Name of the Programme: BTech (Hons.) CSE Name of the Course: Programming in C Semester: I		
Course credit:	No. of hours per week (L+T+P):	Total no. of Teaching hours:
4	3+0+2	45L+30P = 75 Hours
Course Objectives:		
a) Emphasize hands-on coding practices during the initial phase of the course, allowing students to apply theoretical knowledge through practical exercises and problem-solving tasks.		
b) Guide students through key programming constructs such as control statements, loops, and input/output operations, building a deeper understanding of program logic and flow.		
c) Cover intermediate to advanced topics including arrays, functions (with a focus on recursion), pointers, and dynamic memory allocation to prepare students for more complex programming challenges.		
d) Equip students with the skills to manage structured data using structures and unions, and introduce file handling techniques for data storage and retrieval in C programs.		

Syllabus:

Module – 1: Computer Fundamentals in C Language	No. of hours
	3
Fundamentals of Computers: C Language: Structure of C Program, Memory Layout of C Programs, Phases of developing a running computer program in C, how does a C Program Execute: Pre-processor, Compiler, Linker, Loader, Library Functions, Various components of a computer that programs require for execution.	

Module – 2: Data Concepts and Operators	No. of hours
	6
Data concepts in C: Constants, Variables - Variables as named memory locations, Rules for Constructing Variables, understanding different Data Types - range, size, and appropriate usage, C Keywords, format specifiers, Uses and Common Errors associated with them, Instruction, Pre-processor Directive, Macros. Operator in C: Arithmetic Operators - Hierarchy of operations, Precedence, Relational Operators, Logical Operators, Conditional Operators, Bit-wise Operators, Assignment Operators. Type Conversion and Type Casting.	

Module – 3: Decision Control, Loops, and Functions	No. of hours
	14
Decision Control in C: if Statement, else-if Statement, nested else -if Statement - Use case, example for nested if-else, switch. Loop statements: for, while, do-while, break (from loop), continue, exit - exit status to represent the exit code of the program, program clean ups done by exit function. Introduction to Functions: Function Declaration and Definition. Function Prototype. Calling a Function, Return Value. Parameters and Arguments. Local, Global, and Static Variables. Pass by value and reference. Understanding the use of pointers. Recursion: Basic concept of recursion, Study of basic problems using recursion: factorial, GCD, Fibonacci series.	

Module – 4: Arrays and Strings	No. of hours
	12
Arrays: Declaration and Initialization, Indexing, Array Size and Bounds. One-dimensional, Two-dimensional and Multi-dimensional arrays. passing array elements to functions, passing entire array to functions, Basic array operations- traversal, insertion, deletion, updation. Array manipulation- reversing, concatenation, splitting, sorting. Array library functions. String: Strings as array of characters, pointers and strings, string operations. String library functions.	

Module – 5: Structures and File operations	No. of hours
	10
Structures: Introduction to Structures and Unions, Structure Definition and Declaration, Accessing Structure Members. Array of Structures. Passing Structures to Functions. Basic file operations: Creating a new file, opening an existing file, reading from file, writing to a file, moving to a specific location in a file, Closing a file.	

Course Outcomes: After completing the course, the students will be able to:	
a.	Understand and analyse the concepts of the number system, computer memory, and the fundamental concepts of the C programming language.
b.	Demonstrate proficiency in using fundamental data types, operators, and console input/output functions in C
c.	Design and implement C programs using decision-making constructs, loops, arrays, strings, functions, and pointers to solve mathematical and logical problems.
d.	Analyse and evaluate the programming concepts based on user-defined data types and file handling in C language.

Text Books	
1	Let Us C: Authentic guide to C programming language by Yashavant Kanetkar. BPB Publications; Nineteenth edition (15 December 2022); BPB Publications, Ansari Road, Dariya Ganj. ISBN-10: 9355512767. ISBN-13: 978-9355512765.
2	C Programming Absolute Beginner's Guide by Greg Perry and Dean Miller. Que Publishing; 3rd edition (22 August 2013). ISBN-10: 0789751984. ISBN-13: 978-0789751980.
3	Computer Programming in C by Rajaraman V. Prentice Hall India Learning Private Limited (1 January 1994). ISBN-10: 9788120308596. ISBN-13: 978-8120308596

Reference books	
1	How To Solve It By Computer by R G Dromey. ISBN-10: 8120303881. ISBN-13: 978-8120303881
2	Problem Solving and Programming Concepts by Sprankle, Pearson Education India; 9th edition (1 January 2013). ISBN-10: 9789332518841. ISBN-13: 978-9332518841
3	The C Programming Language by Brian W. Kernighan and Dennis Ritchie. Pearson Education India; 2nd edition (1 January 2015). ISBN-10: 9332549443. ISBN-13: 978-9332549449

Lab Programs /Tutorials	
1	Write a program using function-like macros to add two numbers.
2	Write a program to swap two numbers with and without using a third variable.
3	Write a program to create a menu-driven basic arithmetic calculator.
4	a) Write a program to print a Fibonacci series. b) Write a program to compute GCD of two numbers.
5	a) Write a program to print nth term of a Fibonacci series using function. b) Write a program to compute GCD of two numbers using function.
6	a) Write a program to print a Fibonacci series using recursion. b) Write a program to compute GCD of two numbers using recursion.
7	Write a menu driven program to demonstrate the following array operations using functions: Insert, Delete, Update element at index i.
8	Write a program to implement different string operations using functions.
9	Write a program using structures to read, write and compute average-marks of n students. Take student name, USN, and marks in 5 subjects as input. Pass structure by reference to a function to update marks in one of the five subjects as given by user for a particular student. Finally display the new average for that student.
10	Write a program to demonstrate different file operations in C.